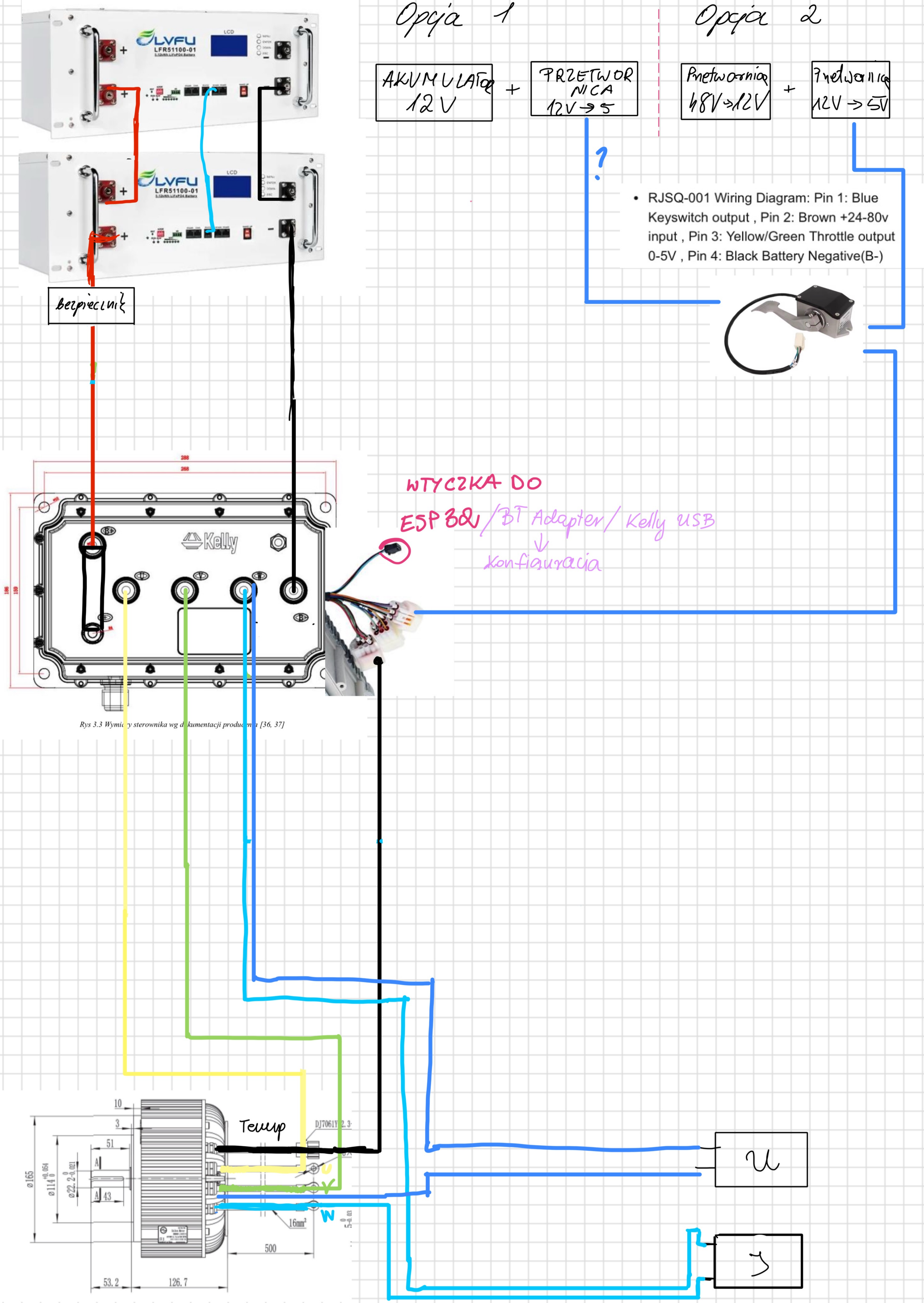


Rys 3.3 Wymiary sterownika wg dokumentacji producenta [36, 37]

3.1. Silnik elektryczny

Tabela 3.1 Specyfikacja silnika elektrycznego według danych producenta

Producent	Golden Motor- BLDC MOTOR
Model	HPM05KW
Napięcie	48V
Moc	5 kW
Chłodzenie	Powietrzem

Tabela 3.2. Tabela specyfikacji sterownika wg danych producenta

Producent	Kelly Controls
Model	KLS12202-8080N
Typ	BLDC z czujnikami Halla (wariant „N”)
Zakres napięcia baterii	24-120 V
Prąd (wg tabeli modelowej)	200 A (60s) / 80 A ciągły
Częstotliwość pracy	10-20 kHz
Zasilanie logiki (PWR)	8-30 V
Zasilanie czujników	5 V / 12 V, prąd zasilania czujników 40mA

3.3 Akumulator

Tabela 3.3. Specyfikacja akumulatora według danych producenta [38, 39]

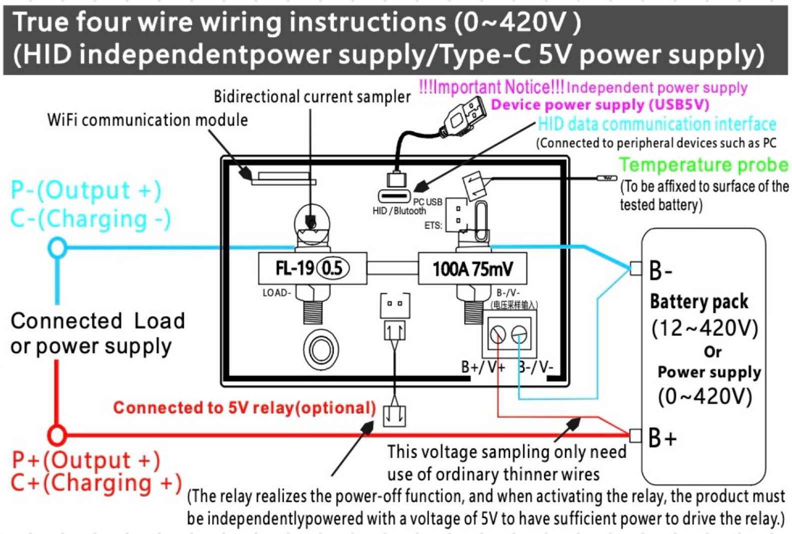
Model	LFR48100
Typ ogniwa	LFP (LiFePO4)
Napięcie nominalne	48 V
Zakres napięcia ładowania	40,5-54 V
Maks. prąd ładowania	50 A
Maks. Prąd rozładownia	100 A
Pojemność znamionowa	100 Ah
Energia znamionowa 4.8 kWh	4.8 kWh
Wymiary (dł. x szer. x wys.)	482.6 x 468 x 178 mm
Masa pojedynczego pakietu	49 ± 3 kg
Interfejsy	RJ45 CAN/485 do komunikacji z inwerterem + RJ45 RS485A/RS485B do łączenia równoległego

Tabela 3.4. Konfiguracja wynikowa

2 Moduły	
Połączenie	Równolegle
Napięcie zestawu	48 V
Pojemność zestawu	200 Ah (*2 x 100Ah*)
Energia zestawu	9.6 kWh (*2x 4,8 kWh*)
Maks. prąd rozładownia	200 A

3.4 **Elektryczny akcelerator**

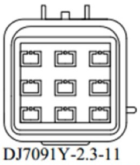
Typ sygnału wejściowego	Analogowy 0-5V
Zakres napięcia zasilania	12V - 48V
Typ czujnika	Potencjometr
Zakres temperatury pracy	-40°C do +85°C



Sterownik

3.2 Connections

3.2.1 Pin definition of KLS-N Controller



DJ7091Y-2.3-11
See from output side

Orange REV-SW (14)	Black GND (6)	White FWD (12)
Red 12V (11)	Yellowish 12V Brake (25)	Blue ECO (22)
Greenish CAN_H (33)	Pink PWR (7)	Brownish CAN_L (34)

DJ7091Y-2.3-21
See from output side

Gray Foot_SW (15)	Green Throttle (3)	
Black GND (20)	D-Gray Meter (8)	
Purple 5V (4)	Brown Brake_AN (2)	Red 12V (11)

DJ7061Y-2.3-21
See from output side

Black GND (21)	Raddle Temp (1)	Purple 5V (5)
Yellow Hall A (18)	D-Green Hall B (17)	D-Blue Hall C (16)

Figure 10: waterproof connector

DJ7091Y-2.3-11 Pin Definition

1. **REV_SW(Pin 14): Reverse switch input. ※Orange**
2. **GND(Pin 6): Signal return or power supply return. ※Black**
3. **FWD(Pin 12): Forward switch or can be enabled as High speed switch function. ※White**
4. **12 V(Pin 11): 12 V Supply. ※Red**
5. **12 V (Pin 25): brake switch. ※Yellowish**
6. **ECO(Pin 22): Low speed switch. ※Blue**
7. **CAN-H(Pin 33): (Optional function). ※Greenish**
8. **PWR(Pin 7): Controller power supply (input). ※Pink**
9. **CAN-L(Pin 34): (Optional function). ※Brownish**

DJ7091Y-2.3-21 Pin Definition

1. **Foot_SW(Pin 15): Throttle switch input. ※Gray**
2. **Throttle(Pin 3): Throttle analog input, 0-5 V. ※ Green**

3. **GND(Pin 20): Signal return. ※Black**
4. **Meter(Pin 8): Copied signal of hall-A sensor. ※Dark Gray**
5. **5 V(Pin 4): 5 V Supply, <40mA. ※Purple**
6. **Brake_AN(Pin 2): Brake variable regen or Boost function. ※Brown**
7. **12 V(Pin 11): 12 V Supply. ※Red**

DJ7061Y-2.3-21 Pin Definition

1. **GND(Pin 21): Signal return. ※Black**
2. **Temp(Pin 1): Motor temperature sensor input. ※Raddle.**
3. **5 V(Pin 5): 5 V Supply, <40mA. ※Purple**
4. **Hall A(Pin 18): Hall sensor signal of phase-A. ※Yellow**
5. **Hall B(Pin 17): Hall sensor signal of phase-B. ※Dark Green**
6. **Hall C(Pin 16): Hall sensor signal of phase-C. ※Dark Blue**

Notes:

1. All GND pins are internally connected.
2. The Meter function outputs the Hall-A sensor signal.
3. By default, the three-gear and three-speed functions cannot be used simultaneously, as both use the same pin (FWD, Pin 12) for control.
4. Switch signals are valid at 12 V.
5. The 12 V output (Pin 11) is intended only for switch signals, with a total current not exceeding 40 mA.
6. CAN bus functionality is not included in KLS-N controllers by default.
7. The Boost and Brake Analog Regen functions share the same port (Brake_AN, Pin 2). When Boost is disabled in the user program, Pin 2 serves as Brake Analog Regen. When Boost is enabled, Pin 2 is used for the Boost function. Due to this port conflict, both functions cannot operate simultaneously on the same port.

3.2.2 KLS-N Controller Standard Wiring

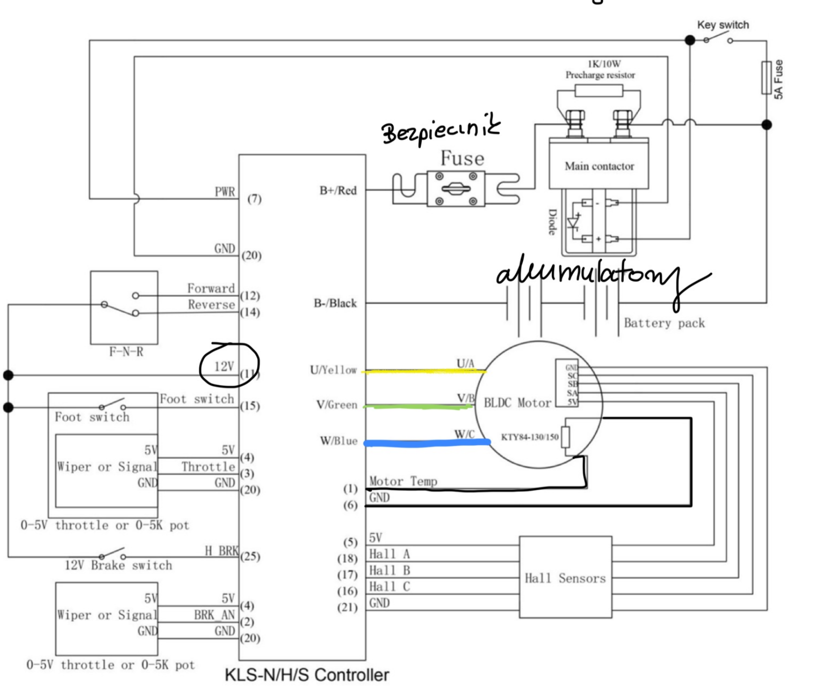


Figure 11: KLS-N controller standard wiring
(Battery is also used as controller's power supply)

Caution!

1 **Verify all controller wiring before powering on.** Double-check connections, especially the B+ and B- terminals. Incorrect wiring can cause severe damage to the controller.
2 Ensure the **B-** connection is secure and properly tightened before applying power.The system contactor or circuit breaker should be connected **in series with the B+ line**.
3 Important: Any contactors installed in the B+ line must have a diode placed across their coils. This diode acts as a freewheeling (flyback) diode to protect the controller's power module.
4 Failure to install this diode may result in serious damage to the controller.
5 Refer to the KLS-N controller standard wiring diagram for proper diode installation.

3.2.3 Communication Port

A 4pin connector is provided to communicate with host for calibration and configuration.

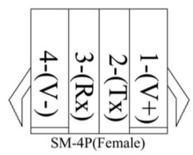


Figure 12: SM-4P connector

3.3 Installation Check List

1. **Conduct a visual inspection** to ensure all components—such as mounting holes, wiring, and sealing rings—are intact and undamaged.
2. **Check the connection between the battery B+ and NC.**
   a) For controllers without a fuse, check the connection between the battery B+ and the controller B+ instead of NC.
3. **Check the connection between the battery B- and the controller B-.** Ensure the connection is secure and properly tightened.
4. **Verify all signal wire connections.** Ensure that PWR and GND are properly isolated from each other.
5. **Inspect the motor Hall wire connections.** Confirm that the 5 V and GND wires align correctly with the motor's interface.
6. **Inspect the throttle wire connections.** Confirm that the 5 V and GND wires align correctly with the throttle's interface.
7. **Check the gear wire connections.** By default, the switch signal is valid at 12 V.

Akumulator

Przekształtnik ???

Ak 12V → 5,5

Przekształtnik

12 V → sterownika

z tyłu

5V → do ESP, wyświetlacz

300mA

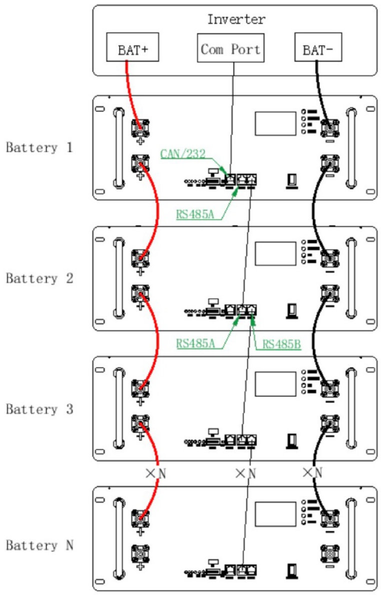


Figure 7. Parallel Connection Diagram

(3) This product supports up to 16 batteries for parallel use

(4) After the installation of the battery system, pay attention to the insulation treatment of the battery poles and cover the insulation cover.

4.4.3 Parallel dialing address selection (manual dialing method)

Definition of parallel DIP switch: In multi machine communication when the battery pack is in parallel, the DIP switch is used to distinguish different pack addresses, and the hardware address can be set through the DIP switch on the board.

Definition of dial switches bit1 to bit8: bit1 to bit4 are used to set the address, and bit5 to bit8 are used for the number of slaves.

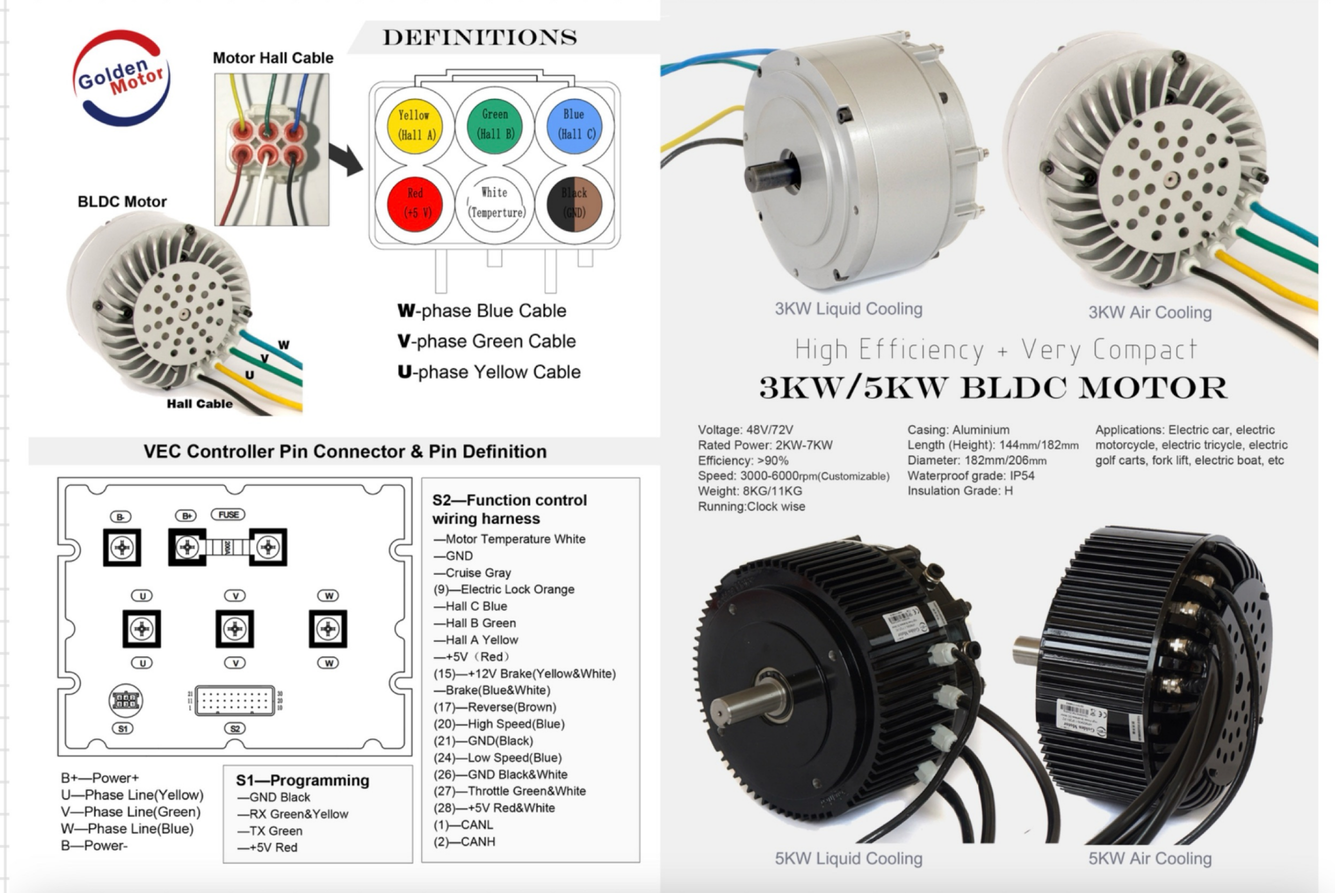
Item	Crystal head picture	Serial no.	Definition
		1	RS485_B
		2	RS 485_A
		3	GND_COM
		4	CANH
		5	CANL
		6	GND_COM
		7	RS 485_A
		8	RS485_B

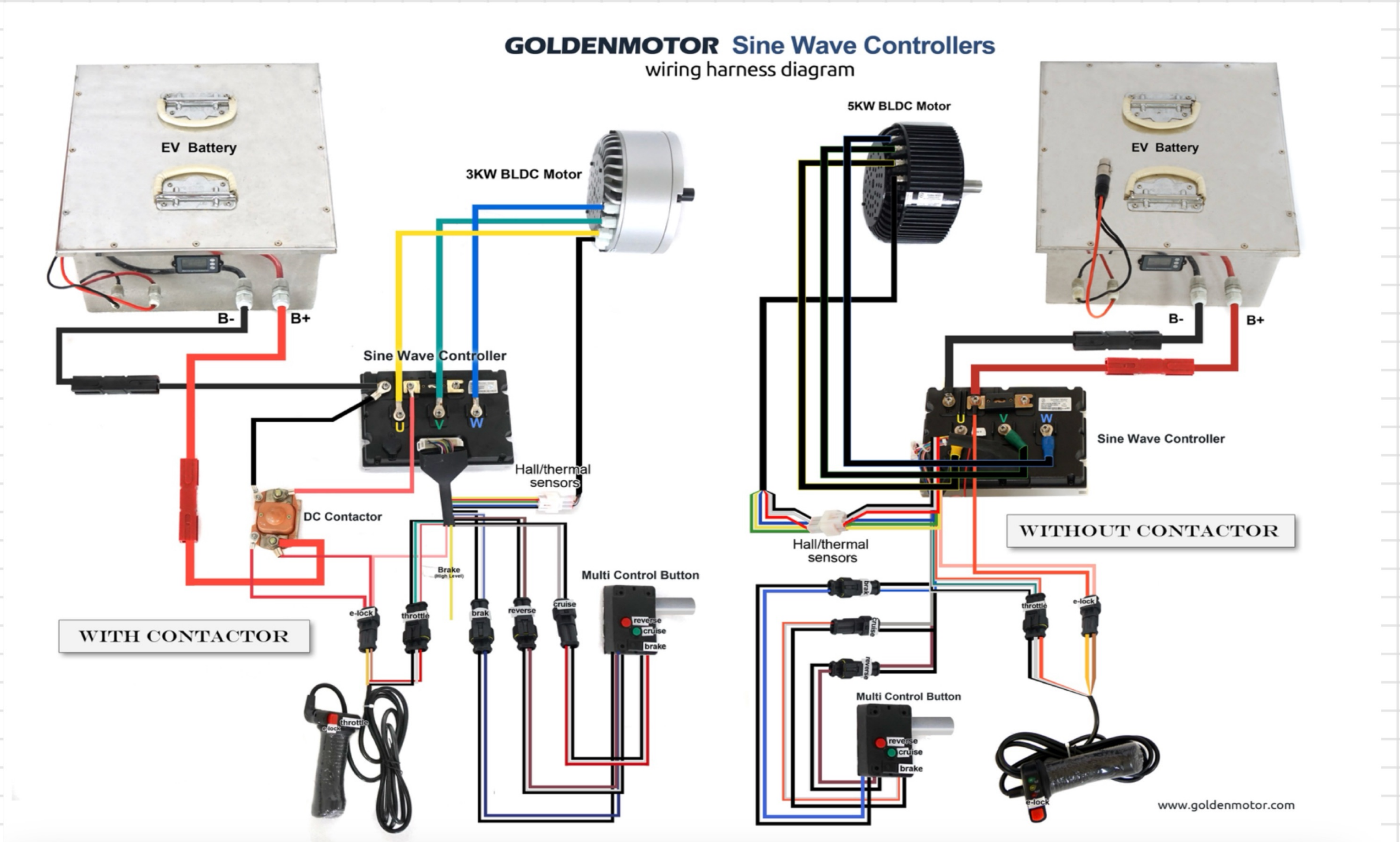
Figure 5. Definition of communication line pins

The definitions of interface 9 (RS485A) and interface 10 (RS485B) are as follows:

NO.	Description
1、8	RS485-B
2、7	RS485-A
3、6	GND
4、5	Internal Communication

Figure 6: Definition of Centralized Communication Interface





Drawing _ --_ → PC AutoCAD